CSE 225 Data Structures
Week #5

A
Quiz #2

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Q1.

Consider the following prefix expression! Build an arithmetic expression tree and write the equivalent postfix and infix expressions!

+ / ^ * a b c + * d e / f ^ g h i

Q2.

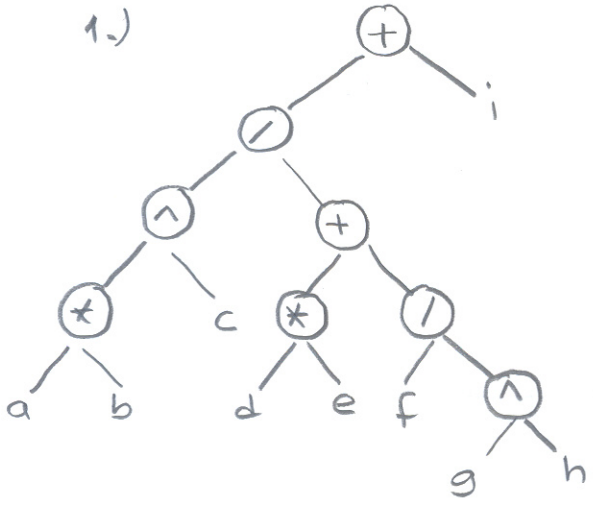
Consider the following infix expression! Build an arithmetic expression tree and write the equivalent postfix and prefix expressions!

(a * b) ^ (c / (d * e) + (f / g) ^ (h + i))

Q3. (f is bonus part of the question!!)

a) Construct a BST from the sequence of keys below!
   14 21 33 86 8 11 3 97 53 68 13 45 79 72
b) Give two more different sequences that give the same BST!
c) What is the best case, worst case, and average case search times of this specific BST!
d) How many depth levels does this BST have?
e) What is the height of the tree?
f) How many different **best** BST topologies (best BST is the BST with the *shortest average* search time for a given set of keys) can you construct from the keys in the above sequence? (Here, consider only the keys! The sequence is not important!)
g) Show one of these best BSTs and give a sequence that provides this BST topology!

1.)

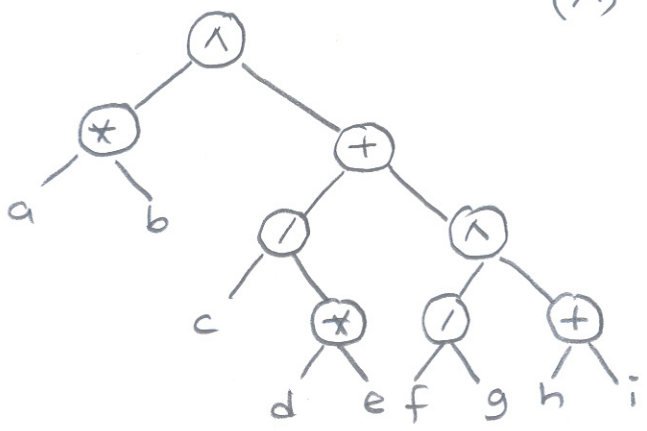


Postfix : ab*c^de*fgh^/+/i+

Infix: (([(a*b)^c] / {(d*e)+[f/(g^h)]})+i

(A)

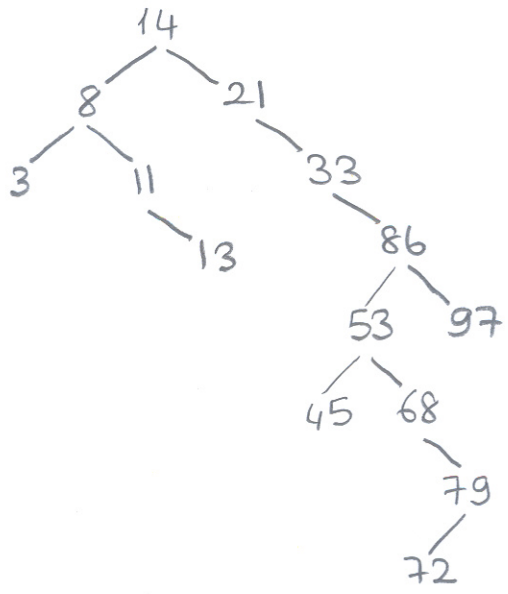
2.)



Postfix: ab*cde*/fg/hi+^+^

Prefix : ^*ab+/c*de^/fg+hi

3.) a)



b) Example sequences can be:

1) 14, 8, 21, 3, 11, 33, 13, 86, 53, 97, 45, 68, 79, 72

2) 14, 21, 8, 11, 3, 13, 33, 86, 53, 97, 45, 68, 79, 72

c) best case = 1

worst case = 8

average case = $\frac{1*1 + 2*2 + 3*3 + 4*2 + 5*2 + 6*2 + 7*1 + 8*1}{14}$

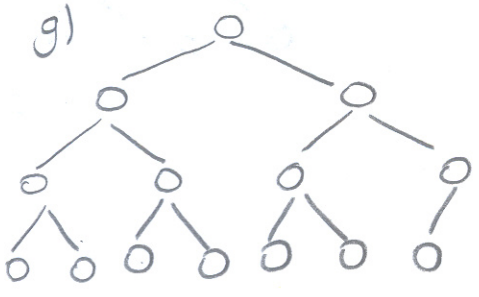
$= 4,21$

d) 8 depth levels (0, 1, ---, 7)

e) Height = 7

f) $\binom{8}{7} = 8$ different best BST topologies.

g)



→ one of the best topologies.

sequence can be:

45, 13, 8, 3, 11, 21, 14, 33, 79, 68, 53, 72, 97, 86